

SEYMOUR JOHNSON AFB, NC, USA

WMO: 723066

Lat: 35.344N

Lon: 77.965W

Elev: 33

StdP: 100.93

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13713

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-3.7	-16.0	0.9	-2.3	-12.9	1.2	0.4	10.8	11.8	9.2	12.1	2.5	280	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	35.8	24.6	34.2	24.5	33.0	24.1	27.0	31.0	26.3	30.5	25.8	29.9	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.1	25.4	20.7	27.6	24.8	19.9	27.3	84.9	31.2	82.0	30.5	79.4	30.1	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.7	6.9	DB	-10.0	37.7	3.0	1.4	-12.1	38.7	-13.9	39.5	-15.5	40.3	-17.7	41.4	
			WB	-10.8	28.1	2.8	1.3	-12.8	29.1	-14.4	29.8	-16.0	30.6	-18.0	31.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.6	8.9	12.1	17.2	21.7	25.8	27.4	26.5	23.5	17.9	11.9	8.5
	DBStd	8.48	6.08	5.66	5.24	4.39	3.80	2.93	2.18	2.30	3.01	4.31	4.74	5.34
	HDD10.0	392	141	81	39	3	0	0	0	0	0	2	33	93
	HDD18.3	1493	366	268	204	73	15	1	0	0	2	61	198	306
	CDD10.0	3081	35	49	103	219	362	474	540	510	404	247	90	48
	CDD18.3	1142	2	3	9	39	119	224	282	252	156	48	6	3
	CDH23.3	10608	6	23	89	357	1062	2235	2883	2361	1172	370	34	14
	CDH26.7	4430	0	2	16	87	392	1029	1353	1031	428	89	1	1
Wind	WSAvg	3.1	3.2	3.4	3.6	3.6	3.2	3.0	2.8	2.6	2.8	2.7	2.8	2.9
Precipitation	PrecAvg	1231	109	90	106	83	94	102	147	140	130	72	74	80
	PrecMax	1802	207	184	192	208	240	278	346	426	775	243	167	188
	PrecMin	878	28	18	18	13	18	22	28	32	2	1	13	15
	PrecStd	189	45	42	44	46	45	57	76	75	126	55	41	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	26.2	28.7	30.8	35.0	37.2	37.7	36.8	34.4	31.7	26.3	25.2
		MCWB	18.9	18.1	17.4	19.7	22.9	24.4	24.9	25.0	22.8	22.1	19.0	18.0
	2%	DB	21.1	23.2	25.9	29.0	32.6	35.3	35.7	34.8	32.6	29.0	24.1	22.2
		MCWB	16.9	16.3	17.3	19.0	22.0	24.4	25.2	24.7	22.8	21.2	17.6	17.9
	5%	DB	18.8	20.7	23.5	27.5	30.8	33.7	34.1	33.1	31.1	27.2	22.2	19.9
		MCWB	15.5	15.1	16.0	18.5	21.5	24.0	25.2	24.8	22.8	20.0	17.2	17.0
	10%	DB	16.3	18.3	21.1	25.3	28.8	32.2	32.8	31.8	29.4	25.2	20.0	17.6
		MCWB	13.3	14.0	15.0	17.9	20.9	23.6	24.8	24.2	22.5	19.6	16.4	15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	19.9	21.1	23.8	24.9	26.9	28.4	27.8	26.5	24.9	21.7	20.7
		MCDB	22.2	23.3	24.0	26.8	28.9	32.1	31.8	31.2	30.0	27.6	23.4	22.8
	2%	WB	18.2	18.5	19.4	21.6	23.9	25.9	27.3	26.8	25.5	23.8	20.2	18.9
		MCDB	20.3	21.9	22.5	25.0	28.5	30.7	31.3	30.5	28.8	26.4	21.9	21.1
	5%	WB	16.4	16.8	18.1	20.5	23.0	25.3	26.6	26.2	24.8	22.7	18.8	17.2
		MCDB	18.3	19.5	21.1	24.3	27.5	30.3	30.9	30.0	27.9	25.1	21.2	19.1
	10%	WB	13.8	14.6	16.8	19.3	22.3	24.7	25.9	25.6	24.0	21.1	17.2	15.3
		MCDB	15.7	17.8	20.3	23.5	26.6	29.6	30.3	29.5	26.8	23.7	19.2	17.4
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.6	12.0	12.3	11.5	10.8	10.2	10.0	9.9	11.8	11.8	10.4
		MCDBR	13.4	13.9	14.9	14.1	13.3	12.4	11.8	11.4	11.9	13.0	13.6	12.8
		MCWBR	9.8	8.9	7.8	6.4	4.9	4.1	3.8	3.8	4.4	6.0	8.1	8.8
	5% WB	MCDBR	11.3	12.8	11.9	11.4	11.1	11.0	10.6	9.9	9.1	9.6	10.9	11.7
		MCWBR	9.9	9.8	8.4	6.8	4.8	4.2	4.3	4.2	4.3	5.7	8.3	9.6
Clear-Sky Solar Irradiance	taub	0.316	0.332	0.361	0.405	0.458	0.503	0.537	0.524	0.423	0.371	0.334	0.317	
	taud	2.494	2.442	2.394	2.272	2.163	2.083	2.002	2.030	2.313	2.433	2.493	2.526	
	Ebn at Noon	888	910	909	880	834	794	763	765	835	854	858	864	
	Edh at Noon	83	98	112	133	151	163	176	168	121	98	83	77	
All-Sky Solar Radiation	RadAvg	2.61	3.30	4.35	5.49	6.04	6.39	6.09	5.49	4.58	3.75	2.88	2.30	
	RadStd	0.17	0.36	0.30	0.36	0.45	0.40	0.35	0.30	0.35	0.47	0.22	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+161	+113

Nomenclature: See separate page